

Calculus of functions of multiple variables

Problem Sheet 3

1. Obtain all the first- and second-order partial derivatives of the following functions $f(x, y)$ and verify $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$ in each case

(i) $xy + \frac{x}{y}$, (ii) $\sin(x^2 y^3)$.

2. If $u(x, y) = x^4 - 6x^2 y^2 + y^4$ and $v(x, y) = 4x^3 y - 4xy^3$, find all first-order derivatives of u and v and verify that

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad (1), \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad (2).$$

Show further that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, by (a) differentiating u directly and (b) differentiating (1) with respect to x and (2) with respect to y and comparing your results.

3. Find the stationary points of the following functions and determine whether they correspond to maxima, minima or saddle points:

(a) $f = x^2 - xy + y^2 + 2x + 2y$,
(b) $f = (2x + 3y - 13)(x - y + 1)$,
(c) $f = x^2 + y^2 - 2x^4$.

4. Show that the function

$$2x^4 - 3x^2 y^2 + y^4 + 8x^2 + 3y^2$$

has stationary values at the points $(0, 0)$, $(5, \pm 6)$, $(-5, \pm 6)$. Identify the nature of these stationary points.

5. The angles of a triangle are inaccurately measured to be $A = 32^\circ$, $B = 64^\circ$, $C = 82^\circ$. The actual angles a , b , c satisfy the relation

$$a + b + c = 180. \quad (1)$$

Find the values of a , b , c which minimise the sum

$$(a - 32)^2 + (b - 64)^2 + (c - 82)^2$$

of the squares of the errors by using (1) to eliminate c from the minimisation.

6. By minimising $x^2 + y^2 + z^2$, the square of the distance of the point (x, y, z) from the origin, find the point on the plane

$$2x + 3y + 4z = 58$$

which is nearest to the origin.

7. **Total Derivatives:** If $f = f(x, y)$ and $x = x(t)$, $y = y(t)$ write down an expression for $\frac{df}{dt}$ using the chain rule.

(a) If $f = x^2 + y^2 + x^2y^2$ and $x = \cos t$, $y = \sin t$, evaluate $\frac{df}{dt}$ using the chain rule.

(b) In (a) find the expression for f as a function of t by substituting for x and y ; hence confirm the form of $\frac{df}{dt}$ found in (a).

8. **Change of Variables:** The temperature T at a point has the functional form $T = T(x, y)$ in terms of the cartesian coordinates (x, y) of the point, and the form $T = \bar{T}(r, \theta)$ in terms of the polar coordinates (r, θ) of the same point, where $x = r \cos \theta$, $y = r \sin \theta$.

Given

$$T(x, y) = x^3 - 3xy^2 \quad (1),$$

confirm that $\bar{T}(r, \theta) = r^3 \cos 3\theta$.

Find the forms of

$$(i) \quad \frac{\partial \bar{T}}{\partial r}, \quad (ii) \quad \frac{\partial \bar{T}}{\partial \theta}$$

by differentiating (1) using the change of variables formula.

Confirm your results by differentiating $\bar{T}(r, \theta)$.

[You may assume $\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$ and $\sin 3\theta = -\sin^3 \theta + 3 \sin \theta \cos^2 \theta$]

9. (a) Given $u = x - 2y^2$ and $v = x + 2y^2$

Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ in forms of $\frac{\partial f}{\partial u}$ and $\frac{\partial f}{\partial v}$.

(b) Show that the partial differential equation

$$\frac{1}{4y} \frac{\partial f}{\partial y} + \frac{\partial f}{\partial x} = f$$

can be converted to a simpler differential equation involving $\frac{\partial f}{\partial v}$ and f .

(c) Solve this and hence give the general solution for $f(x, y)$.

10. Find the stationary points of

$$(x^2 + y^2)^2 + 2a^2(y^2 - x^2),$$

where a is a constant, and identify their nature.

Prerequisites:

$P(n) = \{\text{questions with prerequisite Parts } \leq n\}$

$A = \text{all of PS3.}$

$P(8) = \{1, 2\}$

$P(9) = \{1, 2, 3, 4, 5, 6, 10\} = A \setminus \{7, 8, 9\}$

$P(10) = A$